

$$A_{n \times n}, \det A \neq 0$$

1. Metoda macierzowa

$$X = A^{-1} \cdot B$$

$$A = \begin{bmatrix} x & y & z \\ 1 & -1 & 2 \\ 2 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 9 \\ -2 \end{bmatrix}$$

2. Metoda wyznacznikowa (układ Cramera) $W = \det(A) = \begin{vmatrix} 1 & -1 & 2 \\ 2 & -1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = -1 - 2 + 2 = -1 \neq 0$

$$x = \frac{W_x}{W} \quad y = \frac{W_y}{W} \quad z = \frac{W_z}{W}$$

$$W = \det(A)$$

$$\begin{cases} x - y + 2z = 1 \\ 2x - y = 0 \\ -x \end{cases}$$

$$W_x = \begin{vmatrix} 1 & -1 & 2 \\ 0 & -1 & 0 \\ -2 & 0 & 1 \end{vmatrix} = \dots \quad x = \frac{W_x}{W}$$

$$y = \frac{W_y}{W}$$

$$z = \frac{W_z}{W}$$

$$W_y = \begin{vmatrix} 1 & 1 & 2 \\ 2 & 0 & 0 \\ -1 & -2 & 1 \end{vmatrix} = \dots$$

$$W_z = \begin{vmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ -1 & 0 & -2 \end{vmatrix} = \dots$$

$$\det A = -1$$

$$A^{-1} = \begin{bmatrix} \begin{vmatrix} -1 & 0 \\ 0 & 1 \end{vmatrix} & -\begin{vmatrix} 2 & 0 \\ -1 & 1 \end{vmatrix} & \begin{vmatrix} 2 & -1 \\ -1 & 0 \end{vmatrix} \\ -\begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} & \begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix} & -\begin{vmatrix} 1 & -1 \\ 2 & 0 \end{vmatrix} \\ \begin{vmatrix} -1 & 2 \\ -1 & 0 \end{vmatrix} & -\begin{vmatrix} 1 & 2 \\ 2 & 0 \end{vmatrix} & \begin{vmatrix} 1 & -1 \\ 2 & -1 \end{vmatrix} \end{bmatrix} = \begin{bmatrix} -1 & -2 & -1 \\ 1 & 3 & 1 \\ 2 & 4 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det} \cdot [A^D]^T$$

$$A^{-1} = -1 \cdot \begin{bmatrix} -1 & 1 & 2 \\ -2 & 3 & 4 \\ -1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & -2 \\ 2 & -3 & -4 \\ 1 & -1 & -1 \end{bmatrix}$$

$$X = A^{-1} \cdot B = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 & -1 & -2 \\ 2 & -3 & -4 \\ 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} =$$

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$$3) \begin{cases} 3x + y = 6 \\ -x + 2y = 5 \end{cases}$$

$$W = \det(A) = \begin{vmatrix} 3 & 1 \\ -1 & 2 \end{vmatrix} = 6 + 1 = 7$$

$$A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$$

$$W_x = \begin{vmatrix} 6 & 1 \\ 5 & 2 \end{vmatrix} = 12 - 5 = 7$$

$$B = \begin{bmatrix} 6 \\ 5 \end{bmatrix}$$

$$W_y = \begin{vmatrix} 3 & 6 \\ -1 & 5 \end{vmatrix} = 15 - 6 = 9$$

$$x = 1 \quad y = 3$$

$$A^D = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$$

$$\det A = 7$$

$$X = \begin{bmatrix} \frac{2}{7} & \frac{1}{7} \\ -\frac{1}{7} & \frac{3}{7} \end{bmatrix} \cdot \begin{bmatrix} 6 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad \begin{matrix} x=1 \\ y=3 \end{matrix}$$

$$A^{-1} = \frac{1}{7} \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} \frac{2}{7} & -\frac{1}{7} \\ \frac{1}{7} & \frac{3}{7} \end{bmatrix}$$

	6 5	
$\frac{2}{7} \cdot \frac{1}{7}$	$\frac{12}{7} - \frac{5}{7} = 1$	
$\frac{1}{7} \cdot \frac{3}{7}$	$\frac{6}{7} + \frac{15}{7} = 3$	

$$\begin{cases} 2x+y-z=6 \\ x+2y+z=3 \\ x+2y+2z=1 \end{cases}$$

$$W_x = \begin{vmatrix} 6 & 1 & -1 \\ 3 & 2 & 1 \\ 1 & 2 & 2 \end{vmatrix} = 24 - 6 + 1 + 2 - 12 - 6 = 3 \quad x = \frac{3}{3} = 1$$

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 6 \\ 3 \\ 1 \end{bmatrix}$$

$$W_y = \begin{vmatrix} 2 & 6 & -1 \\ 1 & 3 & 1 \\ 1 & 1 & 2 \end{vmatrix} = 12 - 1 + 6 + 3 - 2 - 12 = 6$$

$$y = \frac{6}{3} = 2$$

$$\det(A) = \begin{vmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ 1 & 2 & 2 \end{vmatrix} = 8 - 2 + 1 + 2 - 4 - 2 = 3$$

$$W_z = \begin{vmatrix} 2 & 1 & 6 \\ 1 & 2 & 3 \\ 1 & 2 & 1 \end{vmatrix} = 4 + 12 + 3 - 12 - 12 - 1 = -6$$

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} & -\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} & \begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix} \\ -\begin{vmatrix} 1 & -1 \\ 2 & 2 \end{vmatrix} & \begin{vmatrix} 2 & -1 \\ 1 & 2 \end{vmatrix} & -\begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} \\ \begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix} & -\begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} \end{bmatrix} = \begin{bmatrix} 2 & -1 & 0 \\ -4 & 5 & -3 \\ 3 & -3 & 3 \end{bmatrix}$$

$$z = \frac{-6}{3} = -2$$

$$A^{-1} = \frac{1}{3} \cdot \begin{bmatrix} 2 & -4 & 3 \\ -1 & 5 & -3 \\ 0 & -3 & 3 \end{bmatrix} =$$

$$\begin{array}{r|l} 2 & 4 & 3 & 12 - 12 + 3 = 3 \\ -1 & 5 & -3 & -6 + 15 - 3 = 6 \\ 0 & -3 & 3 & 0 - 9 + 3 = -6 \end{array}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1} \cdot B = \frac{1}{3} \begin{bmatrix} 2 & -4 & 3 \\ -1 & 5 & -3 \\ 0 & -3 & 3 \end{bmatrix} \cdot \begin{bmatrix} 6 \\ 3 \\ 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3 \\ 6 \\ -6 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}$$

$$x=1 \quad y=2 \quad z=-2$$

$$\begin{cases} x + y + z = 2 \\ x + y - z = 1 \\ x - y + z = 0 \end{cases}$$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

$$\det(A) = \cancel{1} - \cancel{1} - 1 - 1 - 1 - 1 = -4$$

$$A^{\infty} = \begin{bmatrix} \begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix} & -\begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} \\ -\begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} & -\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} \\ \begin{vmatrix} 1 & -1 \\ 1 & -1 \end{vmatrix} & -\begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} \end{bmatrix} = \begin{bmatrix} 0 & -2 & -2 \\ -2 & 0 & 2 \\ -2 & 2 & 0 \end{bmatrix}$$

$$A^{-1} = \frac{1}{-4} \begin{bmatrix} 0 & -2 & -2 \\ -2 & 0 & 2 \\ -2 & 2 & 0 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1} \cdot B = -\frac{1}{4} \begin{bmatrix} 0 & -2 & -2 \\ -2 & 0 & 2 \\ -2 & 2 & 0 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = -\frac{1}{4} \begin{bmatrix} -2 \\ -4 \\ -2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ 1 \\ \frac{1}{2} \end{bmatrix}$$

$$x = \frac{1}{2} \quad y = 1 \quad z = \frac{1}{2}$$

$$2) \quad A = \begin{bmatrix} 2 & 5 \\ 3 & 9 \end{bmatrix} \quad B = \begin{bmatrix} 4 & 0 \\ 6 & 1 \end{bmatrix}$$

$$A \cdot X = B \Rightarrow X = A^{-1} \cdot B$$

$$A^{\textcircled{2}} = \begin{bmatrix} 9 & -3 \\ -5 & 2 \end{bmatrix} \quad \det A = 18 - 15 = 3$$

$$A^{-1} = \frac{1}{\det A} (A^{\textcircled{2}})^T = \frac{1}{3} \begin{bmatrix} 9 & -5 \\ -3 & 2 \end{bmatrix}$$

	4	0
	6	1
9 - 5	6	-5
-3 2	0	2

$$X = A^{-1} \cdot B = \frac{1}{3} \begin{bmatrix} 9 & -5 \\ -3 & 2 \end{bmatrix} \cdot \begin{bmatrix} 4 & 0 \\ 6 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -\frac{5}{3} \\ 0 & \frac{2}{3} \end{bmatrix}$$

Eliminacja Gaussa

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[A|B] \xrightarrow[\text{operacje wektorowe}]{\text{pivot}} [A_I|C]$$

$$\begin{array}{c} \text{pivot} \\ \downarrow \\ \textcircled{2} \end{array} \begin{bmatrix} 2 & 3 & 1 & | & 6 \\ 1 & 2 & -1 & | & 2 \end{bmatrix} \xrightarrow{2w_2 - w_1} \begin{bmatrix} 2 & 3 & 1 & | & 6 \\ 0 & 1 & -3 & | & -2 \end{bmatrix} \xrightarrow{w_1 - 3w_2} \begin{bmatrix} 2 & 0 & 10 & | & 12 \\ 0 & 1 & -3 & | & -2 \end{bmatrix}$$

$$\xrightarrow{\frac{1}{2}w_1} \begin{bmatrix} 1 & 0 & 5 & | & 6 \\ 0 & 1 & -3 & | & -2 \end{bmatrix} \Leftrightarrow \begin{cases} 1 \cdot x + 0 \cdot y + 5 \cdot z = 6 \\ 0 \cdot x + 1 \cdot y - 3 \cdot z = -2 \end{cases} \Rightarrow \begin{cases} x = 6 - 5t \\ y = 3t - 2 \\ z = t \end{cases} \quad t \in \mathbb{R}$$

$\swarrow$ zmienne niezależne (bazowe) $\searrow$ zmienna zależna (PARAMETR)

$$4) \left[\begin{array}{ccc|c} \textcircled{1} & 2 & -1 & 2 \\ 1 & 1 & 1 & 0 \\ 3 & 4 & 1 & 0 \end{array} \right] \xrightarrow[\substack{w_2 - w_1 \\ w_3 - 3w_1}]{} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 2 \\ 0 & -1 & 2 & -2 \\ 0 & -2 & 4 & -6 \end{array} \right] \xrightarrow[\substack{w_1 + 2w_2 \\ w_3 - 2w_2}]{} \left[\begin{array}{ccc|c} 1 & 0 & 3 & -2 \\ 0 & -1 & 2 & -2 \\ 0 & 0 & 0 & -2 \end{array} \right]$$

$$0 \neq -2$$

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$$\left[\begin{array}{ccc|c} \textcircled{1} & 2 & 3 & 1 \\ -1 & -1 & -1 & -2 \\ 0 & 1 & 2 & -1 \end{array} \right] \xrightarrow{w_2 + w_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 1 & 2 & -1 \end{array} \right] \rightarrow$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & -1 \end{array} \right] \xrightarrow{w_1 - 2w_2} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 3 \\ 0 & 1 & 2 & -1 \end{array} \right]$$

$$\begin{cases} x - z = 3 \\ y + 2z = -1 \end{cases}$$

$$\begin{cases} x = t + 3 \\ y = -1 - 2t \\ z = t \end{cases} \quad t \in \mathbb{R}$$

$$\left[\begin{array}{ccc|c} \textcircled{1} & 2 & -1 & 0 \\ 1 & 1 & 1 & 0 \\ 3 & 1 & 1 & 0 \end{array} \right] \xrightarrow[\substack{w_2 - w_1 \\ w_3 - 3w_1}]{} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & -5 & 4 & 0 \end{array} \right] \xrightarrow[\substack{w_1 + 2w_2 \\ w_3 - 5w_2}]{}$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 3 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & -6 & 0 \end{array} \right] \xrightarrow[\substack{w_1 + \frac{1}{2}w_3 \\ w_2 + \frac{1}{3}w_3}]{} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -6 & 0 \end{array} \right] \rightarrow$$

